

Image-to-Image Retrieval System using Classical Computer Vision Techniques

ENCS5343 – Computer Vision

First Project

Diaa Badaha (1210478)

December 27, 2025

1 Introduction

Image-to-image retrieval aims to retrieve images from a large database that are visually similar to a given query image. Unlike image classification, which assigns semantic labels, instance-level image retrieval must remain robust to variations in viewpoint, illumination, scale, and background clutter. This makes the task particularly challenging for landmark datasets, where the same structure may appear under drastically different visual conditions.

Classical image retrieval systems address this problem by relying on local invariant features, visual vocabularies, and similarity-based ranking. Despite the recent dominance of deep learning methods, classical pipelines such as Bag-of-Visual-Words (BoVW) combined with SIFT features remain important both as strong baselines and as educational tools for understanding retrieval fundamentals.

In this project, we design and implement a complete classical image retrieval pipeline and evaluate it on the Oxford5k Buildings dataset. The system is assessed using standard retrieval metrics including Precision@K, Recall@K, mean Average Precision (mAP), and average retrieval time.

2 Proposed Method

Our retrieval system follows a classical Bag-of-Visual-Words (BoVW) pipeline enhanced with TF-IDF weighting and optional dimensionality reduction using Principal Component Analysis (PCA). Figure 1 illustrates the overall system workflow.

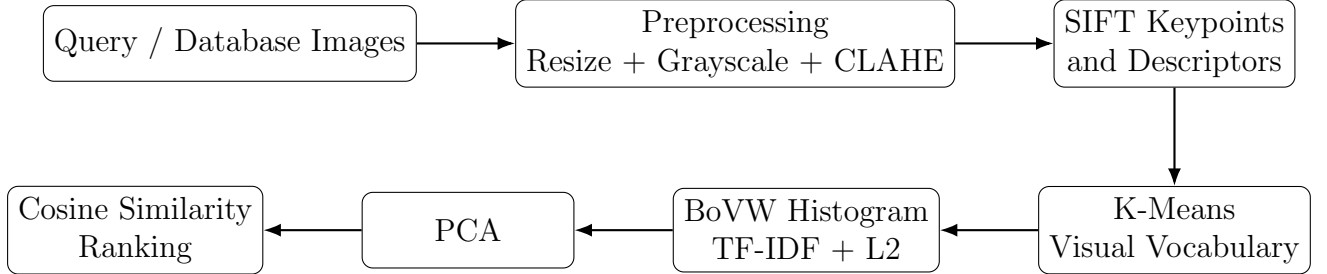


Figure 1: Overview of the proposed image retrieval pipeline.

2.1 Preprocessing

Each image is resized while preserving its aspect ratio such that the maximum dimension does not exceed 1024 pixels. This reduces computational cost while preserving sufficient visual detail. Images are then converted to grayscale, as SIFT descriptors are computed from intensity information.

To improve robustness against illumination changes and low-contrast regions, Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied. CLAHE enhances local contrast without over-amplifying noise, which helps produce more stable local features across varying lighting conditions.

2.2 Local Feature Extraction (SIFT)

Local features are extracted using the Scale-Invariant Feature Transform (SIFT). SIFT detects salient keypoints that are invariant to scale and rotation, and robust to moderate affine and illumination changes. Each keypoint is represented by a 128-dimensional descriptor encoding local gradient information.

To further improve matching performance, RootSIFT normalization is applied by L1-normalizing each descriptor followed by element-wise square rooting. This transformation has been shown to improve retrieval performance by making the descriptor space more suitable for Euclidean distance-based clustering and matching.

2.3 Visual Vocabulary Construction

To aggregate local descriptors into a compact global representation, a visual vocabulary is constructed using K-Means clustering. A subset of SIFT descriptors is sampled from the dataset and clustered into $K = 2000$ clusters, where each cluster center represents a visual word.

The vocabulary size is chosen as a trade-off between discriminative power and computational efficiency. Larger vocabularies offer finer visual distinctions but increase memory usage and retrieval time, while smaller vocabularies may lead to quantization errors.

2.4 BoVW Representation and TF-IDF Weighting

Each image is represented as a Bag-of-Visual-Words (BoVW) histogram by assigning each local descriptor to its nearest visual word. The resulting histogram captures the distribution of visual patterns within the image but does not yet account for their discriminative importance.

To address this, TF-IDF weighting is applied. Term Frequency (TF) reflects how often a visual word appears in an image, while Inverse Document Frequency (IDF) down-weights visual words that appear frequently across many images. This emphasizes distinctive visual structures that are more useful for retrieval. Finally, each TF-IDF vector is L2-normalized to enable cosine similarity computation.

2.5 Dimensionality Reduction using PCA

Although TF-IDF vectors are informative, they are high-dimensional and may contain redundancy. To address this, we apply Principal Component Analysis (PCA) to reduce the dimensionality of TF-IDF vectors to 256 dimensions.

PCA is trained on the database vectors and applied consistently to both database and query representations. After projection, vectors are L2-normalized again to preserve cosine similarity behavior. We evaluate the retrieval system both with and without PCA to assess its impact on accuracy and efficiency.

2.6 Similarity Computation

Given a query image, its representation is compared against all database image vectors using cosine similarity. Since all vectors are L2-normalized, cosine similarity reduces to a dot product operation. Images are ranked in descending order of similarity, and the top-ranked results are returned as retrieval output.

3 Experimental Results

3.1 Dataset and Ground Truth

Experiments are conducted on the Oxford5k Buildings dataset, which contains images of Oxford landmarks captured under varying viewpoints and conditions. The dataset provides an official ground truth for evaluation. Images labeled as *good* or *ok* are considered relevant, while *junk* images are ignored. The query image itself is excluded from the retrieved results.

3.2 Evaluation Metrics

We evaluate retrieval performance using the following metrics:

- **Precision@K (P@K)**: the fraction of relevant images among the top- K retrieved.
- **Recall@K (R@K)**: the fraction of all relevant images retrieved within the top- K .
- **Mean Average Precision (mAP)**: the primary metric, computed as the mean of Average Precision over all queries.
- **Retrieval Time**: average time per query, including feature encoding and ranking.

3.3 Quantitative Results

Table 1 summarizes the retrieval performance for the baseline BoVW+TF-IDF system and the PCA-enhanced variant.

Configuration	mAP	P@5	R@5	P@10	R@10	Time (ms)
BoVW + TF-IDF	0.2837	0.5491	0.1283	0.4545	0.1706	96.73
BoVW + TF-IDF + PCA	0.3508	0.6255	0.1442	0.5182	0.1981	91.94

Table 1: Oxford5k evaluation results over 55 official queries.

3.4 Discussion

The baseline BoVW + TF-IDF system provides a strong classical retrieval baseline, achieving competitive performance on the Oxford5k dataset. Applying PCA consistently improves retrieval accuracy across all metrics while slightly reducing retrieval time. This suggests that PCA removes redundant dimensions and noise from the TF-IDF vectors, resulting in a more compact and discriminative representation.

Notably, PCA improves both early precision (Precision@5 and Precision@10) and overall ranking quality (mAP), indicating that the most relevant images are ranked higher. The reduced dimensionality also leads to faster similarity computation, demonstrating a favorable trade-off between accuracy and efficiency.

These results confirm that dimensionality reduction is an effective enhancement for classical BoVW-based retrieval systems, particularly for large-scale image databases.

3.5 Qualitative Results (Single Query)

In addition to quantitative evaluation over all queries, we present qualitative retrieval results for a representative query image from the Oxford5k dataset. Figure 2 and Figure 3 show the top-10 retrieved images for the query `all_souls_000013.jpg`, with and without PCA.

Ground-truth labels are indicated using colored borders: *green* for **good**, *blue* for **ok**, and *red* for **bad** results.



Figure 2: Single query retrieval without PCA. $AP = 0.1767$, $Precision@5 = 0.60$, $Precision@10 = 0.50$. Several non-relevant images appear among the top results, indicating weaker discrimination.



Figure 3: Single query retrieval with PCA (256D). $AP = 0.3489$, $Precision@5 = 1.00$, $Precision@10 = 0.90$. PCA significantly improves ranking quality, with relevant landmark images consistently appearing at top ranks.

These qualitative results clearly demonstrate the impact of PCA on retrieval performance. Without PCA, visually similar but non-relevant structures are ranked highly, whereas PCA produces a more compact and discriminative representation, leading to improved early precision and overall ranking quality.

4 Comparison with Related Work

Table 2 compares our results with representative Oxford5k results reported in the literature. The comparison focuses on classical local-feature-based methods and one widely-cited deep learning baseline. Reported numbers may differ slightly in evaluation details and experimental setup, and are therefore provided as approximate reference points.

Method	Type	Oxford5k mAP
BoW (large vocabulary)	Classical	~ 0.36
VLAD + SSR	Classical	~ 0.38
Improved Fisher Vector	Classical	~ 0.42
R-MAC (CNN-based)	Deep	~ 0.67
BoVW + TF-IDF	Classical	0.2837
BoVW + TF-IDF + PCA	Classical	0.3508

Table 2: Comparison with published Oxford5k results from related work.

5 Conclusion

In this project, we designed and implemented a complete classical image-to-image retrieval system based on local feature extraction and visual word aggregation. The proposed pipeline combines SIFT keypoints, a Bag-of-Visual-Words representation, TF-IDF weighting, and cosine similarity ranking, with an optional PCA-based dimensionality reduction stage.

Experimental evaluation on the Oxford5k Buildings dataset demonstrates that the baseline BoVW + TF-IDF system provides a strong classical retrieval foundation. Incorporating PCA significantly improves retrieval performance across all evaluation metrics, increasing mean Average Precision (mAP) from 0.2837 to 0.3508, while also slightly reducing average retrieval time. This indicates that PCA effectively removes redundancy and noise from high-dimensional TF-IDF vectors, leading to more compact and discriminative image representations.

Both quantitative results and qualitative single-query analysis confirm that PCA improves early ranking quality, consistently placing relevant images at top positions. This improvement is particularly evident in Precision@K and Average Precision scores, highlighting PCA’s impact on the overall ranking structure rather than isolated top-K results.

Overall, the results validate the effectiveness of classical computer vision techniques for instance-level image retrieval and illustrate how carefully chosen enhancements, such as TF-IDF weighting and dimensionality reduction, can significantly improve performance without increasing computational complexity. Despite the emergence of deep learning-based methods, classical pipelines remain competitive, interpretable, and valuable for understanding the fundamental principles of image retrieval.

References

- [1] R. Arandjelović and A. Zisserman, *All About VLAD*, CVPR 2013.
- [2] G. Tolias, R. Sivic, and H. Jégou, *Particular Object Retrieval with Integral Max-Pooling of CNN Activations*, ICLR 2016.